

**Assignment 2 (25%)**  
**STQD6324 Data Management**  
**SEMESTER 2 2025/2026**

Using the MovieLens 100k Dataset (specifically the `u.user`, `u.data`, and `u.item` files), which can be downloaded from: <https://grouplens.org/datasets/movielens/>, write a Python-based data pipeline using Apache Spark and Cassandra to perform the following analytical tasks:

- i) Calculate the average rating for each movie.
- ii) Identify the top ten movies with the highest average ratings.
- iii) Identify users who have rated at least 50 movies and determine their favourite movie genre based on the genre they rated most frequently.
- iv) Find all users who are less than 20 years old.
- v) Find all users whose occupation is “scientist” and whose age is between 30 and 40 years old.

**Technical Requirements**

Your Jupyter Notebook/script must include the following components:

1. Import the required Python libraries for Apache Spark and Cassandra integration.
2. Load and parse the MovieLens dataset files into HDFS.
3. Create Resilient Distributed Dataset (RDD) objects from the raw dataset.
4. Transform the RDDs into Spark DataFrames.
5. Perform appropriate data cleaning and preprocessing steps where necessary.
6. Execute the required analytical queries using:
  - Apache Spark SQL, and/or
  - Cassandra Query Language (CQL).
7. Write the processed DataFrames into Cassandra keyspaces and tables.
8. Read the Cassandra tables back into Spark DataFrames for validation purposes.
9. Present the outputs clearly using appropriate tables or visualizations where necessary.
10. Provide brief interpretations and discussions for each analytical task.

**Additional Requirements**

- For analytical queries involving aggregation or ranking, display only the top ten results where appropriate.
- Clearly explain your workflow using Markdown cells in the Jupyter Notebook.
- Your code should be properly structured, organized, and commented.
- Students are encouraged to specify:
  - Python version,
  - Apache Spark version,
  - Cassandra version,
  - and any additional libraries/packages used.

**Optional Extension:**

You may additionally attempt the above analytical tasks using HBase and/or MongoDB. This optional component may be considered as added value during evaluation.

**Submission Instructions**

Please submit your assignment through a GitHub repository containing:

- A well-documented Jupyter Notebook (`.ipynb`)
- Clear Markdown explanations
- Required Python scripts (if applicable)
- README instructions for reproducibility
- Relevant outputs, screenshots, or visualizations where appropriate

Ensure that:

- the GitHub repository is accessible,
- all required files are included,
- and the notebook can be executed without major modification.

**Submission Deadline:**

11 June 2026

**Assignmen02 Assessment Rubric**

| <b>Criteria</b>                        | <b>Excellent</b>                                                                                       | <b>Moderate</b>                                                                       | <b>Weak</b>                                                                               | <b>Marks</b> |
|----------------------------------------|--------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|--------------|
| <b>Reproducibility</b>                 | Notebook is fully reproducible with complete execution steps and clear documentation                   | Notebook is mostly reproducible with minor missing steps or explanations              | Notebook is difficult or impossible to reproduce due to missing steps or unclear workflow | <b>5</b>     |
| <b>Interpretation &amp; Discussion</b> | Findings are clearly interpreted, logically discussed, and demonstrate strong analytical understanding | Findings are generally understandable with minor gaps in interpretation or discussion | Findings are unclear, poorly explained, or lack logical interpretation                    | <b>15</b>    |
| <b>GitHub presentation</b>             | Repository is well-structured, neatly organized, professional, and easy to follow                      | Repository is partially organized and reasonably understandable                       | Repository is poorly structured, disorganized, and difficult to navigate                  | <b>5</b>     |